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Selecting appropriate machine learning methods for digital soil mapping☆

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ABSTRACT

Digital soil mapping (DSM) increasingly makes use of machine learning algorithms to identify relationships between soil properties and multiple covariates that can be detected across landscapes. Selecting the appropriate algorithm for model building is critical for optimizing results in the context of the available data. Over the past decade, many studies have tested different machine learning (ML) approaches on a variety of soil data sets. Here, we review the application of some of the most popular ML algorithms for digital soil mapping. Specifically, we compare the strengths and weaknesses of multiple linear regression (MLR), k-nearest neighbors (KNN), support vector regression (SVR), Cubist, random forest (RF), and artificial neural networks (ANN) for DSM. These algorithms were compared on the basis of five factors: (1) quantity of hyperparameters, (2) sample size, (3) covariate selection, (4) learning time, and (5) interpretability of the resulting model. If training time is a limitation, then algorithms that have fewer model parameters and hyperparameters should be considered, e.g., MLR, KNN, SVR, and Cubist. If the data set is large (thousands of samples) and computation time is not an issue, ANN would likely produce the best results. If the data set is small (<100), then Cubist, KNN, RF, and SVR are likely to perform better than ANN and MLR. The uncertainty in predictions produced by Cubist, KNN, RF, and SVR may not decrease with large datasets. When interpretability of the resulting model is important to the user, Cubist, MLR, and RF are more appropriate algorithms as they do not function as "black boxes." There is no one correct approach to produce models for predicting the spatial distribution of soil properties. Nonetheless, some algorithms are more appropriate than others considering the nature of the data and purpose of mapping activity.

1. Introduction

Advancement of statistical approaches and geospatial technologies occurring in the late 20th and early 21st century spurred many new tools for digital soil mapping (DSM) [1–3]. Recently, the statistical approach of data mining in particular has been providing useful tools for DSM. Data mining identifies patterns in datasets via statistical methods and then transforms that information into a perceptible structure for further use [4]. Machine learning (ML), as a method of data analysis and a field of artificial intelligence, uses data mining techniques to learn and build a model. These approaches discover

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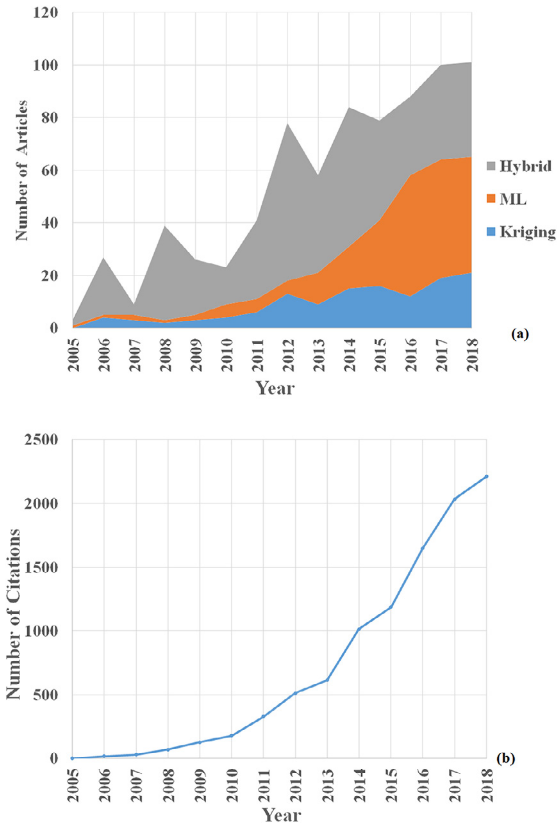

Fig. 1. (a) The quantity of research papers and books on digital soil mapping (DSM) has been increasing and machine learning (ML) approaches have been the largest area of growth within DSM. (b) The growing interest in DSM is also reflected in the increasing quantity of citations for those papers. All data was extracted from a keyword search of “digital soil mapping” [17].

and quantify patterns among data. Most ML techniques can function with minimal human intervention and without being explicitly programmed for the target [5].

Worldwide demand for better performing digital soil maps is growing for three main reasons: (1) limited financial and labor resources to produce soil maps, (2) practical needs for the information these maps provide, and (3) a desire to have quantitative, repeatable results [6,7]. Digital soil maps are becoming important for both scientists and land managers [7] for a wide variety of purposes [8–12]. Although traditional maps utilize the power of human cognition, the uncertainty and efficiency of producing maps based on soil surveyors' expert knowledge is variable between mappers and is difficult to repeat. In this context, the uncertainty refers to how far the best prediction might be from the “true value.” In order to assess uncertainty, two different but equally important validation metrics should be considered: (1) accuracy and (2) variation explained (VE). The qualitative approach to spatial prediction in traditional soil maps also presents a challenge for quantifying this uncertainty. In contrast, digital soil maps model the soil-landscape quantitatively, which provides a more direct basis for measuring uncertainty [13,14]. The quantitative approach to spatial prediction in digital soil maps also offers improved spatial and thematic consistency of information compared to conventional soil maps.

Over the last two decades, the amount of research on DSM has accelerated (Fig. 1). Recently the proportion of those studies utilizing ML has increased, indicating a growing interest for applying popular ML algorithms (i.e., multiple linear regression (MLR), k-nearest neighbors (KNN), support vector regression (SVR), Cubist, random forest (RF), and artificial neural networks (ANN)). In 2005, ANN was the most used ML algorithm for DSM, but since that time research activity with ANN has not increased at the same rate as RF, SVR, and MLR (Fig. 2). Over the past decade, RF has gained the most popularity for DSM, with some interest in KNN, Cubist, and SVR beginning more recently. One of the reasons why decision trees models, e.g., RF and Cubist, are gaining attention is that these algorithms can handle both linear and non-linear relationships of the data. Also, these approaches tend to outperform classical algorithms, e.g., MLR. Another benefit of these types of algorithms is the minimal pre-processing required to run these approaches. Specifically, the data does not need to be transformed [15,16].

Every researcher attempting DSM must decide which approach to use. If that approach will be prediction primarily using relationships with associated covariates, then there is a choice for which algorithm will be used to identify those relationships. In mathematical folklore, there is a theorem popularly referred to as “no free lunch,” meaning no one algorithm

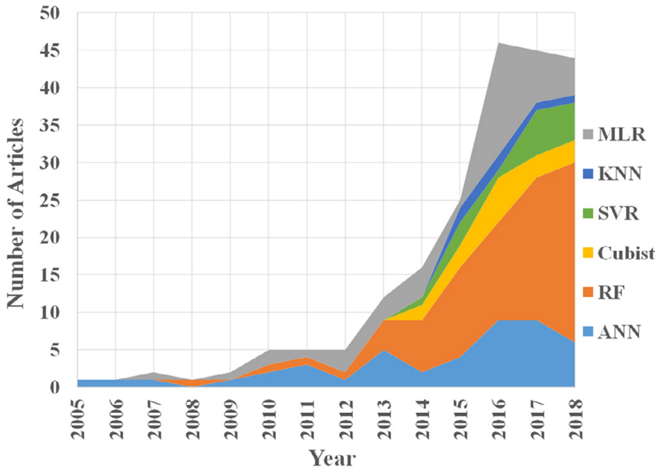

Fig. 2. The number of research papers and books on digital soil mapping (DSM) using popular ML algorithms (MLR, KNN, SVR, Cubist, RF, and ANN) has been increasing. All data was extracted from a keyword search for "digital soil mapping" and the name of algorithm [17].

works best for all problems. High-level programming languages, i.e., R and Python, have increased the accessibility of ML algorithms, leading to their near ubiquitous use in DSM research. Unfortunately, many studies do not provide their rationale for selecting a particular algorithm, e.g., [18–21]. Selecting the most appropriate algorithm for a given set of resource constraints and the desired level of uncertainty depends on the complexity of the algorithm's hyperparameters, quantity of soil samples available, the algorithm's ability to select covariates, efficiency of the computational process, and interpretability of the resulting model.

The soil landscape is complex with multiple environmental factors interacting to influence the direction of processes over time and producing heterogeneous patterns of soil properties [22]. These environmental factors have traditionally been categorized into climate, organisms, relief, and parent material [23]. In this regard, an argument could be made that every soil landscape is a unique combination of soil forming factors. Nonetheless, to advance science, lessons must be extracted from across multiple studies. In addition, ML is also complex with multiple factors influencing outcomes. For a given algorithm, results may vary by the setting of hyperparameters, size of the training sample set, sampling design of the training set, covariates included, and the target variable. While there is currently insufficient data to comprehensively control for all of these factors, this paper attempts to apply known characteristics of ML algorithms [24] to a systematic framework for comparing across DSM studies.

Due to different strengths and weaknesses in the design of each ML algorithm, users would benefit from recognizing the algorithm best suited for their respective mapping conditions. Considering the purpose of the research and the existing limitations of the data can be helpful in selecting the most appropriate ML approach. For example, some algorithms rely on specific assumptions, such as the distribution of the data being normal/Gaussian. In those cases, the data should be transformed to better meet that assumption. In other cases, the data needs to be normalized to the same attribute scale for the algorithm to function properly. However, other algorithms, e.g., decision tree approaches, do not require these specific assumptions. Selecting inappropriate approaches can produce weak or unreliable results. Hence, this paper reviews the characteristics of six ML algorithms that have been the most heavily used in DSM and provides a guide for identifying the most suitable algorithm for the conditions of the mapping activity. While DSM includes the prediction of soil classes and soil properties, this review focuses only on the peer-reviewed papers that studied the prediction of soil properties and are searchable in Scopus. In addition, a section is included on evaluating the results from these ML algorithms in the context of DSM.

2. Selecting the most appropriate algorithms

2.1. Parameters and processes

Model parameters and hyperparameters (a.k.a., tuning parameters) are the variables used to configure a ML algorithm. Model parameters are internal to the model and their values are predicted or learned during training time. Therefore, they are not manually set by the user. Model hyperparameters are external to the model. In other words, users adjust model hyperparameters by trial and error until they reach a minimal amount of error when predictions are examined against the validation dataset. To boost model performance without high error variance and overfitting, users consider tuning options to select an appropriate set of values for the hyperparameters. The calibration of hyperparameters plays a critical role in controlling the training process and producing strong results. However, this calibration is both time consuming and challenging for users. For these reasons, the user's comfort level with the quantity and type of hyperparameters should be a factor in their selection of a ML algorithm.

One of the simplest machine learning methods for finding relationships between several covariates and a target variable is multiple linear regression (MLR) [24,25]. This algorithm is a popular approach to digitally create soil maps [26–28]. With MLR, two or more covariates are used to predict the outcome of the target variable by fitting a linear equation with quantitative coefficients (model parameters). Before performing this fitting, MLR has some assumptions that need to be considered. These assumptions include: (1) linearity (the relationship between the target variable and covariates must be linear), (2) homoscedasticity (the variance of residuals should be similar across the values of covariates), (3) multivariate normality (the residuals are normally distributed), and (4) lack of collinearity (covariates are not correlated with each other). MLR has no hyperparameters to tweak during the training.

K-nearest neighbors (KNN) is also a simple ML algorithm used by soil mappers to create digital maps of soil properties [29,30]. The KNN algorithm predicts the value of a target variable based on the similarity between the target value and its spatial neighbors. This algorithm uses a set of nonparametric regression techniques for predicting a regression curve by learning from all available data. Nonparametric regression models do not require any strong assumptions regarding the regression functions. Therefore, they can use a variety of functional forms for learning from a training dataset. In contrast, parametric models (e.g., MLR) fit the curve on the data with a set of parameters with some assumptions that should be considered. The important hyperparameter for KNN is the quantity of neighbors (k) used to predict the regression curve [31].

Support vector machine/regression (SVM/SVR) is a ML algorithm that has recently gained some popularity in digital soil mapping [32–35]. SVR maintains all covariates to define a maximal margin (margin of tolerance) using the support vectors (observations) and to separate or fit data linearly. The margin is the distance from the decision surface, which is maximally far from any observation. This surface ensures the high generalization ability of the algorithm, making the results more applicable to the unseen data. In addition, this approach applies kernel functions to map non-linear vectors to a very high dimensional space for solving non-linear problems [36]. In the new hyperspace, SVM builds the optimal hyperplane to separate classes, whereas SVR predicts the target variable with minimized empirical risk. Because this paper focuses on the prediction of soil properties – not classification – only SVR will be discussed from this point forward. In terms of hyperparameters, this algorithm requires the user to set the quantity of support vectors and the fraction of support vectors needed to maximize the margin [37].

Cubist is a rule-based algorithm that has recently increased in popularity among digital soil mappers [38–40]. Cubist initially creates a tree structure from a pool of provided covariates. Then the algorithm collapses paths through the tree to create rules using boosting training. Boosting refers to an approach that converts weak learners to strong learners by applying more weight on the stronger learners. Each rule contains an MLR model for predicting the target variable under the conditions of the respective rule. From the top to the bottom of the tree, intermediate nodes smooth the prediction to reduce the prediction error for subsequent nodes based on the prediction obtained at the previous nodes of the tree. The final model located at the terminal nodes shows a collection of MLR models for calculating predicted values [41]. In addition, Cubist as an ensemble model adds boosting to improve the prediction performance using two hyperparameters (i.e., committees and instances) [42]. Ensemble learning combines models produced by multiple repetitions of the same algorithm. This strategy usually obtains stronger predictive performance than results produced from any of the models individually [43].

Random forest (RF) is another algorithm utilizing decision trees and is currently the most commonly used ML algorithm in DSM [18,44,45]. RF uses bootstrap aggregating, called "bagging" to decrease the variance and improve the stability of results. This ensemble learning algorithm randomly selects a group of observations from the training dataset (bootstrapping) and builds a decision tree associated with those selected observations. This process repeats to construct multiple decision trees. RF provides decision trees to produce error predictions using an out-of-bag (OOB) strategy. OOB constructs each tree using different bootstrap sample sets, where two-thirds of the observations are used for training. The remainder of the observations are left out to test model error. RF changes the order of arrangement (permutation) of the covariates randomly and considers all possibilities to select covariates in the OOB samples. Predictions are made based on the mean of results produced from thousands of decision trees. The most important hyperparameters for this algorithm are the number of trees to grow, the number of covariates randomly selected at each node, and the minimum number of sample 'leaves' to capture noise in the training data [46].

Artificial neural networks (ANN), a longstanding and common algorithm in DSM [20,47,48], mimics biological neural networks. ANN constructs a collection of nodes called artificial neurons. Information is transmitted from one neuron to another through multiple layers of the network. Hundreds of artificial neurons or processing elements connect with weights (coefficients) to constitute the network architectures, organize layers, and adjust parameters to learn from the data. The majority of the neural networks are connected from one layer to another. A factor that helps ANN to learn fully is to have a large amount of information which can be provided through the training set. During the training period, the output is compared to the input to compute the residual. The algorithm then goes back through the layers to tweak the network equation and compute the residual again. This process repeats until the smallest residual is achieved. Users can tweak hyperparameters to help ANN decrease the residuals. The hyperparameters include the activation function, the number of hidden layers and hidden neurons, the learning rate (how fast the coefficients of regression and weights of the neural

Table 1
Comparison of studies predicting soil properties with MLR.

Target variable	Sample Size	Sampling Design	Covariate Set	Cross validation		Independent validation		Reference
				R^2	RMSE	R^2	RMSE	
SOC Stock (0–30) (T/ha)	100	Gr	RS+DTA	0.88	1.92	–	–	Roudier et al. [57]
SOC (0–30 cm) (%)	120	StrRd	RS+DTA+L	–	–	0.26	0.87	Mosleh et al. [55]
SOC (0–5 cm) (g/kg)	163	Str	DTA+L	–	–	0.33	13.23	Bonfatti et al. [26]
SOC (5–15 cm) (g/kg)	163	Str	DTA+L	–	–	0.33	13.36	Bonfatti et al. [26]
SOC (15–30 cm) (g/kg)	163	Str	DTA+L	–	–	0.34	12.37	Bonfatti et al. [26]
SOC (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.61	0.49	Mahmoudabadi et al. [56]
SOC stock (kg m $^{-2}$)	139	Gr	DTA+L	0.18	2.26	–	–	Ottoy et al. [86]
SOC (0–30 cm) (g kg^{-1})	184	Str	RS	0.22	6.81	–	–	Campbell et al. [60]
SOC stock (kg m $^{-2}$)	276	Str	DTA+L	0.17	9.56	–	–	Ottoy et al. [87]
SOC (0–30 cm) (%)	334	Str	RS+DTA	0.50	0.35	–	–	Zeraatpisheh et al. [88]
SOC (0–50 cm) (%)	344	PeD	RS+DTA+C	0.28	0.41	–	–	Angelini et al. [28]
SOC (0–200 cm)	595	Str	RS	–	–	0.79	0.52	Dotto et al. [64]
SOC (0–100 cm) (%)	5386	PeD	RS+DTA+C	0.12	0.89	–	–	Somarathna et al. [89]
CEC (0–50 cm) (cmol$^+$ kg^{-1})	344	PeD	RS+DTA+C	0.21	3.23	–	–	Angelini et al. [28]
CEC (50–100) (cmol$^+$ kg^{-1})	344	PeD	RS+DTA+C	0.53	4.04	–	–	Angelini et al. [28]
TN SOC (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.41	0.06	Mahmoudabadi et al. [56]
P (0–30 cm) (mg dm^{-3})	184	Str	RS	0.35	4.26	–	–	Campbell et al. [60]
pH (0–30 cm)	120	StrRd	RS+DTA+L	–	–	0.03	0.15	Mosleh et al. [55]
CCE (0–30 cm) (%)	120	StrRd	RS+DTA	–	–	0.06	11.5	Mosleh et al. [55]
CCE (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.51	9.44	Mahmoudabadi et al. [56]
CCE (0–30 cm) (%)	334	Str	RS+DTA	0.28	9.55	–	–	Zeraatpisheh et al. [88]
EC (0–30 cm) (dS/m)	120	StrRd	RS+DTA+L	–	–	0.11	0.19	Mosleh et al. [55]
BD (0–10 cm) (g cm^{-3})	137	StrRd	RS+DTA	–	–	0.57	0.15	Mahmoudabadi et al. [56]
SMC (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.50	5.9	Mahmoudabadi et al. [56]
Clay (0–30 cm) (%)	120	StrRd	RS+DTA+L	–	–	0.24	8.7	Mosleh et al. [55]
Clay (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.20	11.7	Mahmoudabadi et al. [56]
Clay (0–30 cm) (g kg^{-1})	184	Str	RS	0.4	6.81	–	–	Campbell et al. [60]
Clay (0–30 cm) (%)	334	Str	RS+DTA	0.05	8.33	–	–	Zeraatpisheh et al. [88]
Clay (0–50) (%)	344	PeD	RS+DTA+C	0.19	4.17	–	–	Angelini et al. [28]
Clay (0–20 cm) (g kg^{-1})	399	Str	RS+DTA	–	–	0.53	7.64	da Silva Chagas et al. [27]
Silt (0–30 cm) (%)	120	StrRd	RS+DTA+L	–	–	0.10	12.5	Mosleh et al. [55]
Silt (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.45	4.4	Mahmoudabadi et al. [56]
Silt (0–20 cm) (g kg^{-1})	399	Str	RS+DTA	–	–	0.30	4.94	da Silva Chagas et al. [27]
Sand (0–30 cm) (%)	120	StrRd	RS+DTA+L	–	–	0.05	12.8	Mosleh et al. [55]
Sand (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.40	4.66	Mahmoudabadi et al. [56]
Sand (0–20 cm) (g kg^{-1})	399	Str	RS+DTA	–	–	0.57	9.8	da Silva Chagas et al. [27]

BD, Bulk Density; C, Climate; CCE, Calcium Carbonate Equivalent; CEC, Cation Exchange Capacity; DTA, Digital Terrain Analysis; EC, Electrical Conductivity; Gr, Grid; L, Legacy; P, Phosphorus; PeD, Pre-existing Data; RS, Remote Sensing; SMC, Saturated Moisture Capacity; SOC, Soil Organic Carbon; Str, Stratified; StrRd, Stratified Random; TN, Total Nitrogen.

network change), training type, epoch (the number of iterations), minimum error, and momentum (speed of convergence for the gradient descent learning algorithm) [49].

2.2. Sample size

In general, the quantity of data points plays a critical role in the strength of ML results because larger sample sizes can supply a more accurate prediction of the mean and identify outliers that skew the data. In addition, increasing the sample size decreases the variance of prediction, which tends to improve the model performance. However, the response in model performance to the quantity of training data is different between ML algorithms. Also, sample size is not the same as sampling design. Optimal sampling design is beyond the scope of this paper. Instead the focus is on the capabilities of the ML algorithms to utilize the data in the provided data set. Some ML algorithms, e.g., Cubist and RF, are less sensitive to the quantity of sample points [50], while the model performance of ANN is vulnerable to small sample sizes [51,52]. On the other hand, the processing time for KNN [53], and SVR [54] increases exponentially with sample size.

A meta-analysis of previous digital soil mapping studies suggests model performance generally improves with an increase in the sample size for modeling with MLR (Table 1). Even though sample size is not the only factor affecting R^2 when comparing different studies, a notable pattern can be observed. For example, comparing two studies predicting calcium carbonate equivalent (CCE) using MLR, the R^2 of independent validation improved from 0.06 with 120 observations [55] to 0.51 with 137 observation [56]. In those same studies, the R^2 for independent validation increased from 0.05 to 0.40 for predicting sand content. While that small increase in sample size was unlikely fully responsible for the full increase in R^2, the pattern continues. Also predicting sand content with MLR and 339 samples, de Silva Chagas et al. [27] achieved a R^2 of 0.57 in independent validation. The same pattern is seen for other soil properties. Mosleh et al. [55] predicted soil organic

Table 2
Comparison of studies predicting soil properties with KNN or SVR.

Target variable	Sample Size	Sampling Design	Covariate Set	Cross validation		Independent validation		Reference
				R^2	RMSE	R^2	RMSE	
KNN								
SOC (0–15 cm) (kg m^{-3})	188	Str	RS+DTA	–	0.09	–	–	Taghizadeh et al. [30]
SOC (0–15 cm) (g kg^{-1})	538	Str	DTA+C	0.2	–	–	–	Mansuy et al. [29]
TN (0–15 cm) (g kg^{-1})	538	Str	DTA+C	0.05	–	–	–	Mansuy et al. [29]
BD (0–15 cm) (g cm^{-3})	538	Str	DTA+C	0.17	–	–	–	Mansuy et al. [29]
Clay (0–15 cm) (%)	538	Str	DTA+C	0.43	–	–	–	Mansuy et al. [29]
Silt (0–15 cm) (%)	538	Str	DTA+C	0.13	–	–	–	Mansuy et al. [29]
Sand (0–15 cm) (%)	538	Str	DTA+C	0.19	–	–	–	Mansuy et al. [29]
SVR								
SOC stock (kg m^{-2})	139	Gr	DTA+L	0.27	2.24	–	–	Ottoy et al. [86]
SOC (0–30 cm) (g kg^{-1})	184	Str	RS	0.45	9.13	–	–	Campbell et al. [60]
SOC stock (0–30 cm) (mg ha^{-1})	220	StrRd	RS+DTA+C	–	–	0.64	14.88	Were et al. [20]
SOC stock (kg m^{-2})	276	Str	DTA+L	0.20	9.22	–	–	Ottoy et al. [87]
SOC (0–200 cm)	595	Str	RS	–	–	0.75	0.56	Dotto et al. [64]
SOC Stock (0–5 cm) (mg ha^{-1})	705	Rd	RS+DTA+C	0.40	3.3	–	–	Wang et al. [61]
SOC Stock (0–30 cm) (mg ha^{-1})	705	Rd	RS+DTA+C	0.45	9.2	–	–	Wang et al. [61]
SOC (0–100 cm) (%)	5386	PeD	RS+DTA+C	0.20	0.86	–	–	Somarathna et al. [89]
P (0–30 cm) (mg dm^{-3})	184	Str	RS	0.09	5.89	–	–	Campbell et al. [60]
Clay (0–30 cm) (g kg^{-1})	184	Str	RS	0.25	9.13	–	–	Campbell et al. [60]

BD, Bulk Density; C, Climate; DTA, Digital Terrain Analysis; Gr, Grid; L, Legacy; P, Phosphorus; PeD, Pre-existing Data; Rd, Random; RS, Remote Sensing; SOC, Soil Organic Carbon; Str, Stratified; StrRd, Stratified Random; TN, Total Nitrogen.

carbon (SOC) with an R^2 of 0.26 for independent validation. Bonfatti et al. [26] predicted SOC with 43 more samples than Mosleh et al. [55] and achieved a higher R^2 for independent validation of 0.33, with MLR.

Comparison of previous studies does not indicate a pattern of different sample sizes affecting prediction performance for KNN, SVR (Table 2), Cubist (Table 3), or RF (Table 4). For example, Roudier et al. [57] used Cubist to obtain an R^2 of cross validation (CV) equal to 0.71 for predicting SOC with 100 samples. Zeraatpisheh et al. [14] predicted SOC with the same algorithm and obtained a slightly lower R^2 of CV (0.55), even though the sample size was three times larger. Miller et al. [58] predicted silt content with Cubist and 177 sample points, achieving a R^2 of 0.62 (CV), while Viscarra Rossel et al. [59] only reached a R^2 value of 0.46 (CV) with a set of 14,227 samples and a wider variety of covariates.

In the case of RF, Campbell et al. [60] and Wang et al. [61] predicted SOC (topsoil) with a large difference in sample size, 184 and 705, respectively. However, the accuracy of the model tested by R^2 of CV remained in the 0.44–0.55 range. De Silva Chagas et al. [27] tested RF with independent validation, predicting SOC (topsoil) with a sample size of 399, resulting in a R^2 of 0.56. Vaysse and Lagacherie [62] predicted the same target variable with a sample size of 1766 but still only achieved an R^2 of 0.59 for independent validation.

The major limitation of using ANN is its sensitivity to sample size. Although some ANN studies have achieved strong results, DSM studies have generally not had the large quantity of samples required to obtain stable results with ANN. Stable results have similar model performance between repeated runs of the algorithm. In contrast, unstable results would sometimes produce models with strong performance metrics and sometimes not. Table 5 illustrates high volatility in validation, even when the sample sizes are similar. For example, Mosleh et al. [55] and Mahmoudabadi et al. [56] predicted topsoil SOC with ANN and similar covariates, using 120 and 137 sample points, respectively. However, the R^2 for independent validation ranged from 0.03 to 0.77. Were et al. [20] also used ANN with a similar covariate set to predict topsoil SOC, but with a sample size of 220 only achieved an independent validation R^2 of 0.61. This volatility means that when there is not a massive sample size, i.e., thousands or millions of samples, the ANN algorithm does not reliably produce models that predict soil properties well [63]. Nonetheless, in some studies, ANN was able to create a strong prediction model from smaller sample sizes. For example, R^2 values for independent validation as high as 0.79 was obtained for SOC (0–100 cm) with 595 points [64] and 0.85 for soil moisture content (topsoil) with 137 points [56].

To demonstrate the patterns observed in the above studies, all six algorithms were applied to a matching set of covariates and sample points. Then the quantity of sample points provided to the algorithms was systematically reduced to evaluate the effect on the k-fold cross-validation (10-fold CV) of the resulting models. The target variable was SOC (top 5 cm) sampled using a stratified sampling design across the conterminous USA [65]. The covariate pool included latitude, longitude, and elevation as well as slope gradient, profile curvature, and relative elevation calculated at a 30-m analysis scale. The terrain derivatives were calculated in ArcGIS and the ML algorithms were run using the caret (Classification And REgression Training) package in the R environment. The caret package contains a set of functions, including several ML algorithms, for generating predictive models.

The R^2 for three of the algorithms (MLR, SVR, and ANN) increased, while the other three (KNN, Cubist, and RF) decreased with a decrease in sample size (Fig. 3). In regression analysis, an increase in R^2 with fewer sample points to train on usually indicates overfitting. Therefore, this test suggests that MLR, SVR, and ANN are more susceptible to overfitting. Supporting

Table 3
Comparison of studies predicting soil properties with Cubist.

Target variable	Sample Size	Sampling Design	Covariate Set	Cross validation		Independent validation		Reference
				R^2	RMSE	R^2	RMSE	
SOC (0–75 cm) (g kg^{-1})	49	StrRd	RS+DTA+L	–	–	0.12	12.64	Lacoste et al. [67]
SOC Stock (0–30 cm) (T/ha)	100	Gr	RS+DTA	0.71	2.88	–	–	Roudier et al. [57]
SOC (topsoil) (%)	117	Rd	RS+DTA+L	0.61	–	–	–	Miller et al. [38]
SOC (topsoil) (%)	117	Rd	RS+DTA+L	0.05	–	–	–	Miller et al. [58]
SOC (0–20 cm) (%)	328	PeD	RS+DTA+L	–	–	0.59	2.8	Peng et al. [90]
SOC (0–20 cm) (%)	328	PeD	RS+DTA+L	–	–	0.46	3.6	Peng et al. [90]
SOC (0–30 cm) (%)	334	Str	RS+DTA	0.55	0.34	–	–	Zeraatpisheh et al. [88]
SOC (0–5 cm) (g kg^{-1})	711	Rd	RS+DTA+C + L	0.32	8.24	–	–	Akpa et al. [39]
SOC (5–15 cm) (g kg^{-1})	711	Rd	RS+DTA+C + L	0.26	7.32	–	–	Akpa et al. [39]
SOC (15–30 cm) (g kg^{-1})	711	Rd	RS+DTA+C + L	0.16	5.9	–	–	Akpa et al. [39]
SOC (0–100 cm) (%)	5386	PeD	RS+DTA+C	0.16	0.87	–	–	Somarathna et al. [89]
SOC (0–5 cm) (g kg^{-1})	1994	Rd	DTA+C + L	–	–	0.41	0.24	Adhikari et al. [85]
SOC (5–15 cm) (g kg^{-1})	1994	Rd	DTA+C + L	–	–	0.42	0.24	Adhikari et al. [85]
SOC (15–30 cm) (g kg^{-1})	1994	Rd	DTA+C + L	–	–	0.43	0.66	Adhikari et al. [85]
Zn (0–30 cm) (mg kg^{-1})	300	Str	RS+DTA+L	–	–	12.18	0.51	Peng et al. [91]
Cu (0–30 cm) (mg kg^{-1})	300	Str	RS+DTA+L	–	–	2.1	0.74	Peng et al. [91]
Cr (0–30 cm) (mg kg^{-1})	300	Str	RS+DTA+L	–	–	5.9	0.53	Peng et al. [91]
Pb (0–30 cm) (mg kg^{-1})	300	Str	RS+DTA+L	–	–	1.49	0.53	Peng et al. [91]
Ni (0–30 cm) (mg kg^{-1})	300	Str	RS+DTA+L	–	–	5.19	0.47	Peng et al. [91]
As (0–30 cm) (mg kg^{-1})	300	Str	RS+DTA+L	–	–	0.58	0.6	Peng et al. [91]
Peat Thickness (0–20 cm) (m)	159	Str	RS+DTA	–	–	0.63	2.7	Rudiyanto et al. [40]
BD (topsoil) (g cm^{-3})	117	Rd	RS+DTA+L	0.26	–	–	–	Miller et al. [38]
BD (topsoil) (g cm^{-3})	117	Rd	RS+DTA+L	0.31	–	–	–	Miller et al. [58]
BD (0–5 cm) (mg m^{-3})	222	Rd	RS+DTA+C + L	0.2	0.16	–	–	Akpa et al. [39]
BD (5–15 cm) (mg m^{-3})	222	Rd	RS+DTA+C + L	0.24	0.15	–	–	Akpa et al. [39]
BD (15–30 cm) (mg m^{-3})	222	Rd	RS+DTA+C + L	0.32	0.14	–	–	Akpa et al. [39]
CCE (0–30 cm) (%)	334	Str	RS+DTA	0.3	9.52	–	–	Zeraatpisheh et al. [88]
Clay (0–30 cm) (%)	334	Str	RS+DTA	0.1	8.27	–	–	Zeraatpisheh et al. [88]
Clay (0–5 cm) (%)	14,227	PeD	RS+DTA+C	0.53	11.69	–	–	Viscarra Rossel et al. [59]
Clay (5–15 cm) (%)	14,095	PeD	RS+DTA+C	0.52	12.13	–	–	Viscarra Rossel et al. [59]
Clay (15–30 cm) (%)	12,970	PeD	RS+DTA+C	0.47	13.7	–	–	Viscarra Rossel et al. [59]
Silt (topsoil) (%)	117	Rd	RS+DTA+L	0.83	–	–	–	Miller et al. [58]
Silt (0–5 cm) (%)	14,227	PeD	RS+DTA+C	0.46	0.98	–	–	Viscarra Rossel et al. [59]
Silt (5–15 cm) (%)	14,095	PeD	RS+DTA+C	0.47	0.96	–	–	Viscarra Rossel et al. [59]
Silt (15–30 cm) (%)	12,970	PeD	RS+DTA+C	0.48	0.95	–	–	Viscarra Rossel et al. [59]
Sand (0–5 cm) (%)	14,227	PeD	RS+DTA+C	0.49	16.22	–	–	Viscarra Rossel et al. [59]
Sand (5–15 cm) (%)	14,095	PeD	RS+DTA+C	0.48	16.35	–	–	Viscarra Rossel et al. [59]
Sand (15–30 cm) (%)	12,970	PeD	RS+DTA+C	0.45	17.14	–	–	Viscarra Rossel et al. [59]
SK (topsoil) (%)	117	Rd	RS+DTA+L	0.08	–	–	–	Miller et al. [38]
Soil thickness (topsoil) (cm)	117	Rd	RS+DTA+L	0.12	–	–	–	Miller et al. [38]
Soil thickness (topsoil) (cm)	117	Rd	RS+DTA+L	0.12	–	–	–	Miller et al. [38]

As, Arsenic; BD, Bulk Density; C, Climate; CCE, Calcium Carbonate Equivalent; Cr, Chromium; Cu, Copper; DTA, Digital Terrain Analysis; Gr, Grid; L, Legacy; Ni, Nickel; Pb, Lead; PeD, Pre-existing Data; Rd, Random; RS, Remote Sensing; SK, particles > 2 mm (%); SOC, Soil Organic Carbon; Str, Stratified; StrRd, Stratified Random; Zn, Zinc.

this point, the RMSE increased with decreasing sample sizes for all of the algorithms. RMSE would be expected to increase with decreasing R^2. However, when the R^2 and the RMSE increase together, then the R^2 is more likely to be inflated due to overfitting. An intriguing exception that occurred for all algorithms is that the RMSE abruptly decreased for the smallest sample size of 50, which was most likely due to an extreme amount of overfitting to the small quantity of points.

2.3. Covariate pools

Although there is no universal rule about reducing uncertainty in model predictions by increasing the number of covariates in the pool presented to a ML algorithm, the odds of having the best covariates available to the algorithm increases with a broader selection. Nonetheless, there are no guarantees that the best covariates are included in the pool. For example, Angelini et al. [28] and Zeraatpisheh et al. [14] used MLR to predict SOC with a similar number of observations. Even though Angelini et al. [28] covered a broader selection of covariates by also including climate in the covariate pool, the R^2 of CV for their study was only 0.28, compared to 0.50 for Zeraatpisheh et al. [14]. Using ANN, Mahmoudabadi et al. [56] predicted total nitrogen with only slightly higher accuracy than Kalambukattu et al. [48]; R^2 of independent validation was 0.67 and 0.62, respectively. This difference between the R^2 values was negligible considering Mahmoudabadi et al. [56] used nearly twice as many sample points and included digital terrain derivatives in addition to remote sensing in the covariate pool.

Table 4
Comparison of studies predicting soil properties with RF.

Target variable	Sample Size	Sampling Design	Covariate Set	Cross validation		Independent validation		Reference
				R^2	RMSE	R^2	RMSE	
SOC Stock (0–30) (T ha^{-1})	100	Gr	RS+DTA	0.88	1.83	–	–	Roudier et al. [57]
SOC (0–30) (%)	116	Str	RS+DTA+L	–	–	0.23	0.20	Dharumarajan et al. [92]
SOC (0–30 cm) (g kg^{-1})	184	Str	RS	0.53	6.52	–	–	Campbell et al. [60]
SOC stock (0–30 cm) (mg ha^{-1})	220	StrRd	RS+DTA+C	–	–	0.53	17.57	Were et al. [20]
SOC (0–30 cm) (%)	334	Str	RS+DTA	0.55	0.33	–	–	Zeraatpisheh et al. [88]
SOC (0–200 cm)	595	Str	RS	–	–	0.67	0.66	Dotto et al. [64]
SOC Stock (0–5 cm) (mg ha^{-1})	705	Rd	RS+DTA+C	0.44	3.2	–	–	Wang et al. [61]
SOC Stock (0–30 cm) (mg ha^{-1})	705	Rd	RS+DTA+C	0.45	8.8	–	–	Wang et al. [61]
SOC (0–5 cm) (g kg^{-1})	711	Rd	RS+DTA+C + L	0.34	7.96	–	–	Akpa et al. [45]
SOC (5–15 cm) (g kg^{-1})	711	Rd	RS+DTA+C + L	0.30	6.71	–	–	Akpa et al. [45]
SOC (15–30 cm) (g kg^{-1})	711	Rd	RS+DTA+C + L	0.20	5.57	–	–	Akpa et al. [45]
SOC (0–5 cm) (g kg^{-1})	1766	PeD	RS+DTA+C	–	–	0.59	1.83	Vaysse and Lagacherie [62]
SOC (5–15 cm) (g kg^{-1})	1731	PeD	RS+DTA+C	–	–	0.59	1.85	Vaysse and Lagacherie [62]
SOC (15–30 cm) (g kg^{-1})	1494	PeD	RS+DTA+C	–	–	0.53	1.96	Vaysse and Lagacherie [62]
CEC (0–5 cm) (cmol$^+$ kg^{-1})	996	PeD	RS+DTA+C	–	–	–	9.49	Vaysse and Lagacherie [62]
CEC (5–15 cm) (cmol$^+$ kg^{-1})	995	PeD	RS+DTA+C	–	–	–	9.46	Vaysse and Lagacherie [62]
CEC (15–30 cm) (cmol$^+$ kg^{-1})	966	PeD	RS+DTA+C	–	–	–	9.16	Vaysse and Lagacherie [62]
pH (0–30 cm)	116	Str	RS+DTA+L	–	–	0.3	0.75	Dharumarajan et al. [92]
pH (0–30 cm)	460	Gr	RS+DTA	–	–	–	0.45	Pahlavan-Rad and Akbarimoghaddam [93]
pH (0–5 cm)	1804	PeD	RS+DTA+C	–	–	0.78	0.73	Vaysse and Lagacherie [62]
pH (5–15 cm)	1796	PeD	RS+DTA+C	–	–	0.78	0.73	Vaysse and Lagacherie [62]
pH (15–30 cm)	1740	PeD	RS+DTA+C	–	–	0.76	0.75	Vaysse and Lagacherie [62]
EC (0–30 cm) (ds m^{-1})	116	Str	RS+DTA+L	–	–	0.62	0.14	Dharumarajan et al. [92]
P (0–30) (mg dm^{-3})	184	Str	RS	0.34	4.68	–	–	Campbell et al. [60]
CCE (0–30 cm) (%)	334	Str	RS+DTA	0.23	9.96	–	–	Zeraatpisheh et al. [88]
Peat Thickness (0–20 cm) (m)	159	Str	RS+DTA	–	–	0.61	2.8	Rudiyanto et al. [40]
BD (0–5 cm) (mg m^{-3})	222	Rd	RS+DTA+C + L	0.25	0.16	–	–	Akpa et al. [39]
BD (5–15 cm) (mg m^{-3})	222	Rd	RS+DTA+C + L	0.29	0.14	–	–	Akpa et al. [39]
BD (15–30 cm) (mg m^{-3})	222	Rd	RS+DTA+C + L	0.36	0.13	–	–	Akpa et al. [39]
Clay (0–30 cm) (g kg^{-1})	184	Str	RS	0.39	6.52	–	–	Campbell et al. [60]
Clay (0–20 cm) (%)	321	PeD	RS+DTA	0.68	8.7	–	–	Behrens et al. [71]
Clay (0–30 cm) (%)	334	Str	RS+DTA	0.15	7.86	–	–	Zeraatpisheh et al. [88]
Clay (0–20 cm) (g kg^{-1})	399	Str	RS+DTA	–	–	0.56	7.39	da Silva Chagas et al. [27]
Clay (0–30 cm) (%)	460	Gr	RS+DTA	–	–	–	6.02	Pahlavan-Rad and Akbarimoghaddam [93]
Clay (0–5 cm) (g kg^{-1})	1955	PeD	RS+DTA+C	–	–	0.34	9.99	Vaysse and Lagacherie [62]
Clay (5–15 cm) (g kg^{-1})	1945	PeD	RS+DTA+C	–	–	0.35	9.97	Vaysse and Lagacherie [62]
Clay (15–30 cm) (g kg^{-1})	1889	PeD	RS+DTA+C	–	–	0.31	10.03	Vaysse and Lagacherie [62]
Clay (0–5 cm) (%)	4568	PeD	RS+DTA+C	–	–	0.89	6.48	Akpa et al. [45]
Clay (5–15 cm) (%)	4568	PeD	RS+DTA+C	–	–	0.91	5.71	Akpa et al. [45]
Clay (15–30 cm) (%)	4568	PeD	RS+DTA+C	–	–	0.89	6.93	Akpa et al. [45]
Clay (30–60 cm) (%)	4568	PeD	RS+DTA+C	–	–	0.72	10.04	Akpa et al. [45]
Silt (0–20 cm) (%)	342	PeD	RS+DTA	0.63	10.6	–	–	Behrens et al. [71]
Silt (0–20 cm) (g kg^{-1})	399	Str	RS+DTA	–	–	0.25	5.02	da Silva Chagas et al. [27]
Silt (0–30 cm) (%)	460	Gr	RS+DTA	–	–	–	17.45	Pahlavan-Rad and Akbarimoghaddam [93]
Silt (0–5 cm) (g kg^{-1})	1894	PeD	RS+DTA+C	–	–	0.27	9.76	Vaysse and Lagacherie [62]
Silt (5–15 cm) (g kg^{-1})	1883	PeD	RS+DTA+C	–	–	0.27	9.82	Vaysse and Lagacherie [62]
Silt (15–30 cm) (g kg^{-1})	1830	PeD	RS+DTA+C	–	–	0.29	9.36	Vaysse and Lagacherie [62]
Silt (0–5) (%)	4568	PeD	RS+DTA+C	–	–	0.88	4.44	Akpa et al. [45]
Silt (5–15) (%)	4568	PeD	RS+DTA+C	–	–	0.9	4.14	Akpa et al. [45]
Silt (15–30) (%)	4568	PeD	RS+DTA+C	–	–	0.91	3.43	Akpa et al. [45]
Sand (0–20 cm) (g kg^{-1})	399	Str	RS+DTA	–	–	0.63	9.08	da Silva Chagas et al. [27]
Sand (0–30 cm) (%)	460	Gr	RS+DTA	–	–	–	21.4	Pahlavan-Rad and Akbarimoghaddam [93]
Sand (0–5 cm) (g kg^{-1})	1924	PeD	RS+DTA+C	–	–	0.35	13.98	Vaysse and Lagacherie [62]
Sand (5–15 cm) (g kg^{-1})	1913	PeD	RS+DTA+C	–	–	0.35	14.02	Vaysse and Lagacherie [62]
Sand (15–30 cm) (g kg^{-1})	1863	PeD	RS+DTA+C	–	–	0.35	14.02	Vaysse and Lagacherie [62]
Sand (0–5) (%)	4568	PeD	RS+DTA+C	–	–	0.91	7.7	Akpa et al. [45]
Sand (5–15) (%)	4568	PeD	RS+DTA+C	–	–	0.92	7.05	Akpa et al. [45]
Sand (15–30) (%)	4568	PeD	RS+DTA+C	–	–	0.92	7.26	Akpa et al. [45]

(continued on next page)

Table 4 (*continued*)

Target variable	Sample Size	Sampling Design	Covariate Set	Cross validation		Independent validation		Reference
				R^2	RMSE	R^2	RMSE	
CF (0–5 cm) (%)	1566	PeD	RS+DTA+C	–	–	0.04	9.4	Vaysse and Lagacherie [62]
CF (5–15 cm) (%)	1559	PeD	RS+DTA+C	–	–	0.09	9.28	Vaysse and Lagacherie [62]
CF (15–30 cm) (%)	1492	PeD	RS+DTA+C	–	–	0.17	10.22	Vaysse and Lagacherie [62]

BD, Bulk Density; C, Climate; CCE, Calcium Carbonate Equivalent; CEC, Cation Exchange Capacity; CF, Coarse fragment; DTA, Digital Terrain Analysis; EC, Electrical Conductivity; Gr, Grid; L, Legacy; P, Phosphorus; PeD, Pre-existing Data; Rd, Random; RS, Remote Sensing; SOC, Soil Organic Carbon; Str, Stratified; StrRd, Stratified Random.

Table 5
Comparison of studies predicting soil properties with ANN.

Target variable	Sample Size	Sampling Design	Covariate Set	Cross validation		Independent validation		Reference
				R^2	RMSE	R^2	RMSE	
SOC (0–30 cm) (%)	120	StrRd	RS+DTA+L	–	–	0.03	1	Mosleh et al. [55]
SOC (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.77	0.27	Mahmoudabadi et al. [56]
SOC stock (kg m $^{-2}$)	139	Gr	DTA+L	0.19	2.58	–	–	Ottoy et al. [86]
SOC (0–30 cm) (g kg^{-1})	184	Str	RS	0.31	7.31	–	–	Campbell et al. [60]
SOC stock (0–30 cm) (mg ha^{-1})	220	StrRd	RS+DTA+C	–	–	0.61	15.46	Were et al. [20]
SOC stock (kg m $^{-2}$)	276	Str	DTA+L	0.18	9.79	–	–	Ottoy et al. [87]
SOC (0–200 cm)	595	Str	RS	–	–	0.79	0.52	Dotto et al. [64]
SOC (0–80 cm) (%)	1134	Gr	DTA+C + L	0.8	17.1	–	–	Aitkenhead and Coull [3]
TN (0–15 cm) (%)	75	Gr	RS+DTA	–	–	0.62	0.0006	Kalambukattu et al. [48]
TN (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.67	0.03	Mahmoudabadi et al. [56]
BD (0–10 cm) (g cm^{-3})	137	StrRd	RS+DTA	–	–	0.69	0.08	Mahmoudabadi et al. [56]
BD (0–80 cm) (g cm^{-3})	1134	Gr	DTA+C + L	0.78	0.22	–	–	Aitkenhead and Coull [3]
SMC (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.85	3.4	Mahmoudabadi et al. [56]
Profile depth (cm)	1134	Gr	DTA+C + L	0.58	22.27	–	–	Aitkenhead and Coull [3]
P (0–15 cm)	75	Gr	RS+DTA	–	–	0.51	0.036	Kalambukattu et al. [48]
P (0–30 cm) (mg dm^{-1})	184	Str	RS	0.28	4.5	–	–	Campbell et al. [60]
pH (0–30 cm)	120	StrRd	RS+DTA+L	–	–	0.04	0.15	Mosleh et al. [55]
CCE (0–30 cm) (%)	120	StrRd	RS+DTA+L	–	–	0.25	11.6	Mosleh et al. [55]
CCE (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.72	7.46	Mahmoudabadi et al. [56]
EC (0–30 cm) (dS/m)	120	StrRd	RS+DTA+L	–	–	0.09	0.2	Mosleh et al. [55]
Clay (0–30 cm) (%)	120	StrRd	RS+DTA+L	–	–	0.22	8.7	Mosleh et al. [55]
Clay (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.36	4.46	Mahmoudabadi et al. [56]
Clay (0–30 cm) (g kg^{-1})	184	Str	RS	0.21	7.31	–	–	Campbell et al. [60]
Silt (0–30 cm) (%)	120	StrRd	RS+DTA	–	–	0.65	13.1	Mosleh et al. [55]
Silt (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.7	3.55	Mahmoudabadi et al. [56]
Sand (0–30 cm) (%)	120	StrRd	RS+DTA+L	–	–	0.01	12.9	Mosleh et al. [55]
Sand (0–10 cm) (%)	137	StrRd	RS+DTA	–	–	0.62	5.6	Mahmoudabadi et al. [56]

BD, Bulk Density; CCE, Calcium Carbonate Equivalent; DTA, Digital Terrain Analysis; EC, Electrical Conductivity; Gr, Grid; L, Legacy; P, Phosphorus; RS, Remote Sensing; SMC, Saturated Moisture Capacity; SOC, Soil Organic Carbon; Str, Stratified; StrRd, Stratified Random; TN, Total Nitrogen.

Some algorithms (e.g., Cubist and RF) do not use all the covariates presented to predict the target variable. The reasons for this selection include avoiding the risk of overfitting, decreasing computation time, and for simplification of models to make the result more interpretable. For example, Cubist removes some of the redundant covariates for simplicity and decreasing training time. In doing so, the results are more interpretable as they are less complex. However, in some cases, Cubist removes all covariates, leaving only a constant. In those cases, the constant is usually the mean of the target variable, indicating that none of the covariates helped explain the variability in the target variable. Although this elimination causes an increase in the residual mean, Cubist finally selects the model with the lowest predicted error [41]. RF also does not consider all covariates at the same time. Instead it randomly selects covariates from the full pool of covariates, predicts the target variable, and finally averages the predictions from all decision trees [46].

2.4. Computation time

The time to complete a modeling process consists of multiple steps, including analyzing the problem, testing for conformity of the data with the algorithm's assumptions, adjusting hyperparameters, model building (training), validation of results, interpreting results, and drawing conclusions. This modeling process may be repeated multiple times in order to tweak different elements or conduct experiments. A prime example is the adjustment of hyperparameters through a process of trial and error. An iterative modeling process becomes more sensitive to computing time because the time is multiplied by the number of iterations. Absolute computing time is dependent on the computer hardware available. However, relative

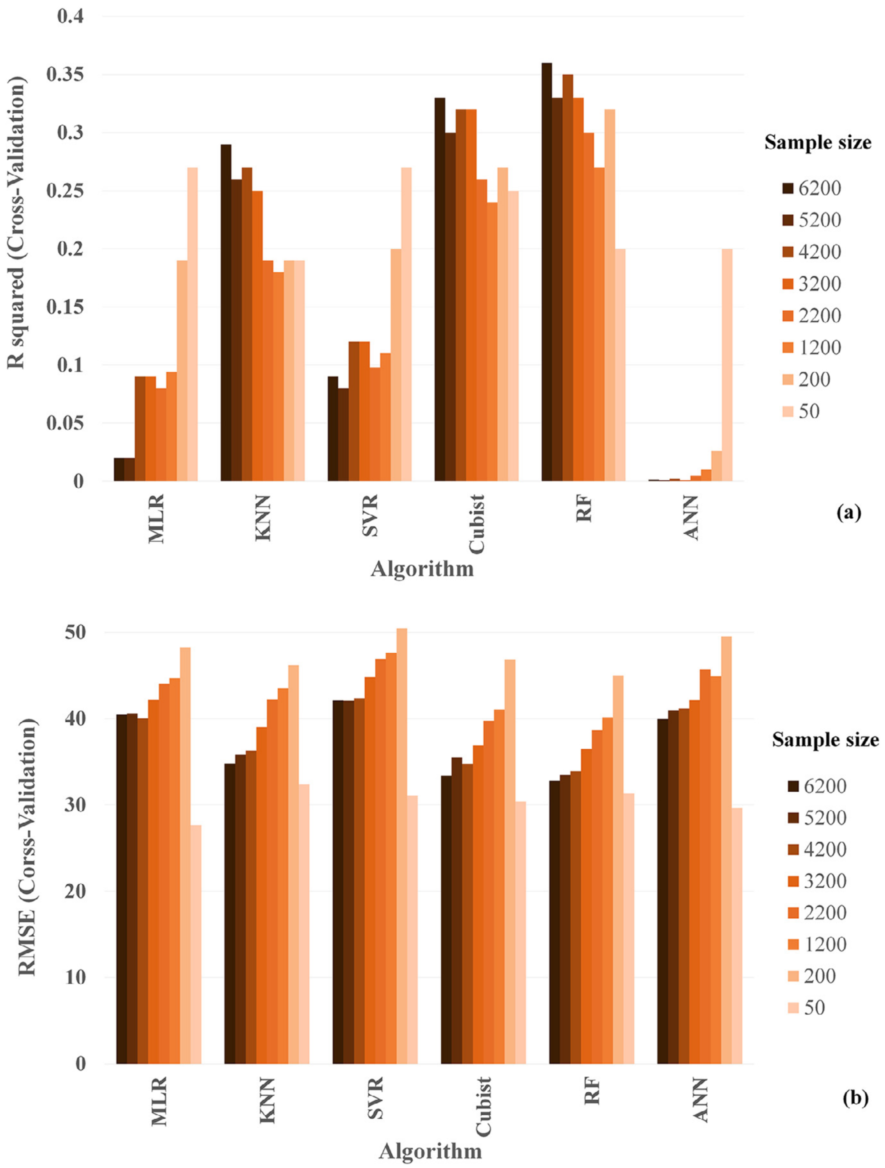

Fig. 3. Case study showing the variation of (a) R^2 and (b) RMSE for different ML algorithms with decreases in sample size. The covariates used to predict soil organic carbon were latitude, longitude, and elevation along with slope gradient, profile curvature, and relative elevation calculated at a 30 m analysis scale across the conterminous USA. Note that the R^2 increased with decreasing sample size for MLR, SVR, and ANN, which may be due to overfitting. An indicator for that overfitting may be seen in the respective increases of RMSE. For the respective ML algorithms, the RMSE tended to increase with decreasing sample sizes. However, there was a remarkable decline in RMSE for the sample size of 50 for all ML algorithms, which could be due to extreme overfitting.

computing time could be influential in the choice of a ML algorithm if multiple iterations will be part of the modeling process.

For the same example data set used in the section on sample size, a test case was run for the computation time to complete hyperparameter calibration and model building for the six ML algorithms. The caret package in *R* was again used for the modeling process. Important in this test, the caret package includes functions for automatically optimizing hyperparameters. Although our experience has been that results can be improved by manually calibrating hyperparameters, using caret to calibrate the hyperparameters helped to standardize the time taken to complete this step.

In this test case, the rate of increased training time with sample size grew from simpler to more complicated algorithms (Fig. 4). For example, the difference in training time between the largest (6200) and the smallest (50) sample size was less than one minute with MLR and KNN. Likewise, this difference in training time was small for SVR and Cubist, taking between one and two minutes. On the other hand, RF and especially ANN were more sensitive to changing sample size for training. With ANN, the training time increased from one minute to one hour when the sample size was increased from 50 to 6200.

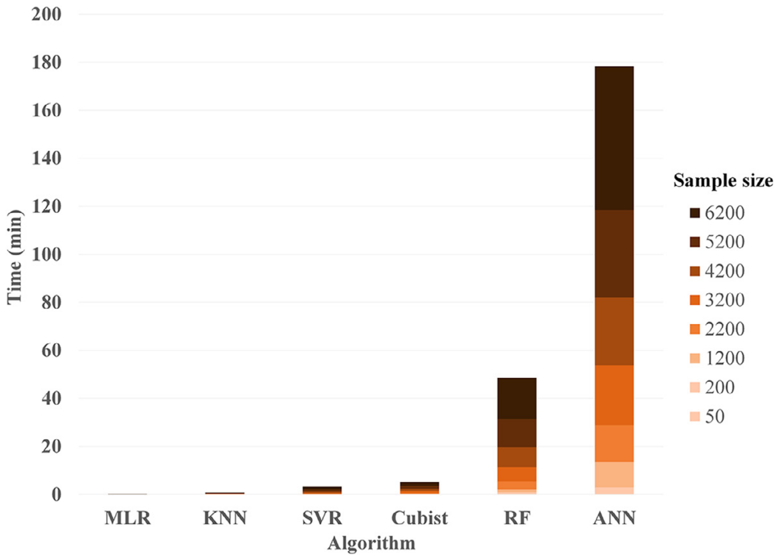

Fig. 4. Graph illustrating the sensitivity of computation time to sample size for different algorithms. The same example data set from Fig. 3 was used in this test. Note that the computation time of ANN and RF have stronger responses to changing sample size, while the other algorithms do not have major differences.

Increasing the sample size for some ML algorithms (i.e., MLR, KNN, SVR, and Cubist) by a magnitude of a hundred, at most only resulted in a doubling of training time. In contrast, that same increase in sample size for RF or ANN increased the time for model training by multiples of 10 and 60, respectively.

2.5. Interpretability

A correct prediction only partially resolves the problem of understanding soil variations. While an accurate prediction is useful, the prediction alone does not explain why the prediction is made. Knowing “the why” helps users to discover potential connections to process as well as distinguish between real and spurious correlations. Interpretability is a gradient, with no model being completely uninterpretable and caution needed when attempting to associate processes with even simple regression models. Nonetheless, different levels of interpretability can be suitable depending on the goals of the modeling activity.

While caution must be used in interpreting individual covariate’s relationship with the target variable, MLR identifies the strength of the combined covariates to predict the target variable. Within an MLR, coefficients of different covariates are difficult to compare directly, because the covariates directly (presence of collinearity) and indirectly (the nature of data) influence the coefficients. If the covariates have not been normalized, then the coefficients are even less useful as the magnitudes of the coefficients are compensating for the different attribute scales of the covariates. Hence, the covariate that is more important in predicting the target variable cannot be identified by simply examining the MLR equation. While the model produced by MLR is a clear, single equation, the model must be interpreted as a whole and not dissected into its component parts.

Cubist produces rules to segment different covariates’ relationships with the target variable. Under each rule is an MLR equation, which has the same limitation in interpreting direct relationships between covariates and the target variable as standard MLR. If no committees are used, a single model is produced showing exactly how predictions were made. The model is more complex than the single equation of MLR and the covariates used in the rules are interpretable in terms of where the relationships between covariates and the target variable are changing. Using committees (boosting) might improve the model performance but may also make the interpretability of the results produced more difficult. To help in the interpretation of the model, Cubist summarizes covariates’ frequency of use in rules and MLR equations. This approach provides some additional insight about the structure of the model and its dependency on certain covariates [39,66–68].

Some studies have considered RF as a “black box” approach [30,44], because the results do not provide clear descriptions of how the predictions are made [67,70]. However, this algorithm does supply some abilities to interpret the model by providing measures for variable importance [71,72]. Trees generated in RF are not themselves interpretable because they are used as an ensemble with other trees to produce predictions. Instead, a summary of the frequency of use for the covariates is calculated after training a RF model and reported as a measure of the respective variable’s importance. This description of importance is not the same as a sensitivity analysis and equally strong models could be produced if the ML algorithm is run again without a covariate that was previously used in high frequency. Nonetheless, the frequency of use provides useful

Table 6
Comparison of different metrics for goodness of fit with different sample sizes and variance values. This demonstration uses the same covariate pool and target variable data as Fig. 3. A single MLR model was produced for a set of 200 points (Dataset A) and then the character of that fitting was parsed to evaluate how that model fit different portions of that training set. Dataset B was a randomly selected subset of dataset A but containing only 50 points. Dataset C is a subset of Dataset B, consisting of the 25 samples with the largest residuals. Dataset D is also a subset of Dataset B, consisting of the other 25 samples with the smallest residuals. Metrics that show the most differentiation are the most sensitive to the effect of sample size and magnitude of residuals. RMSE can show the effect of these factors better than the other metrics.

Dataset	Random Sample Size	MAE	RMSE	MBE	Variance of error	RPD	RPIQ	CCC
A	200	73.58	87.22	74	2242	0.64	0.69	0.14
B	50	21.8	28	0.00	800	1.01	1.23	0.33
C	25	25.6	35.1	8.02	1214	0.99	1.22	0.30
D	25	17.7	18.4	−8.02	286	0.92	1.54	0.46

CCC, Concordance Correlation Coefficient; MAE, Mean Absolute Error; MBE, Mean Bias Error; RPD, Residual Predictive Deviation; RMSE, Root Mean Squared Error; RPIQ, Ratio of Performance to Interquartile.

information about the structure of the model. Therefore, instead of being classified as a black box algorithm, RF may be better described as a semi-black box approach.

In contrast, ANN, KNN, and SVR are true examples of black boxes because the resulting models usually do not make apparent how the covariates and target variable are associated. If users intend to investigate the relationship between covariates and a target variable, black box algorithms would not provide useful information [69,70]. Even inquiring into the code would not provide helpful information about the model produced.

3. Variable selection prior to applying machine learning

Each ML algorithm has unique limitations that can potentially reduce its utility in DSM. Some of these limitations can be mitigated; however, some cannot. If users cannot find a solution to remedy a limitation that is important to their modeling goals, then another type of algorithm should be considered. This section suggests a potential remedy to diminish the effect of certain ML algorithms' limitations, i.e., longer training time and overfitting.

Long training times for some ML algorithms can be too inefficient for some soil mapping objectives. One of the major factors that increases the computation time is high quantities of covariates. Besides the obvious solution of increasing computer power, pre-selecting covariates before running the ML algorithm would decrease training time without sacrificing the type of algorithm or using as many sample points as possible. Variable selection – as the process of selecting important and non-redundant variables – can concentrate the potential covariate pool the ML algorithm needs to process [73]. Although more covariates could help a ML algorithm produce stronger results, a large pool of covariates can increase training time remarkably. Some popular variable selection methods include weight decay [73], LASSO (least absolute shrinkage and selection operator) regression [74], and stepwise regression [75].

Overfitting is a concern because it produces misleading results in ML [76]. Overfitting frequently occurs when either the number of covariates increases, or the sample size decreases. When this happens, the results may show high R^2 and sometimes even low RMSE values, but the model is less likely to be stable or robust for new data. Once a ML algorithm is working hard to find the patterns in a training dataset and it has a large pool of variables to work with, the likelihood of identifying patterns by random chance rather than by true properties of the unseen functions increases. For this reason, the difference between the performance metrics for the training (goodness of fit) versus for the validation set will tend to be large. Trimming down the potential covariate pool with a variable selection tool prior to the ML process can minimize the potential for high quantities of covariates to be overfitted.

4. Evaluation of model predictions

4.1. Metrics of uncertainty

The typical purpose of using a ML algorithm in DSM is to build models with spatial predictive power in order to produce a soil map. Virtually all soil maps are an exercise in spatial prediction. Therefore, a common evaluation of a ML algorithm is the resulting model's ability to predict soil variation. Many approaches are available for evaluating model performance, but they evaluate different aspects of the results. For DSM to advance, some common metrics of model assessment should be used in every research paper to facilitate comparison between studies and the identification of broader trends. While the present paper attempts to compare across DSM studies using ML algorithms, Tables 1–6 also illustrate the challenges in building a greater body of knowledge by connecting independent research activities.

Uncertainty, which is inclusive of accuracy and variation explained (VE), can be measured with a wide range of statistical metrics. The common measures for a model's performance include mean absolute error (MAE), root mean squared error (RMSE), variance of error, mean bias error (MBE), coefficient of determination (R^2), residual predictive deviation (RPD), ratio of performance to interquartile distance (RPIQ), and concordance correlation coefficient (CCC). Each of these metrics describe the differences between the observed and predicted values, or residuals, in different ways.

4.1.1. Accuracy metrics

MAE, RMSE, and variance of error describe the accuracy of the model, which means how close the predicted values are to the actual values. All three of these metrics summarize the residuals. MAE is calculated by averaging the absolute values of the residuals (Eq. (1)). RMSE uses the square root of the sum of the squares of the residuals (Eq. (2)). The variance of error shows the deviation of residuals from the mean of the residuals (Eq. (3)).

$$\mathrm{MAE} = \frac{1}{n}\sum_{j=1}^{n}\left|y_j - \hat{y}_j\right| \tag{1}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{j=1}^{n}\left(y_j - \hat{y}_j\right)^2} \tag{2}$$

$$\text{Variance of error} = \frac{\sum_{j=1}^{n}\left(x_j - \bar{x}\right)^2}{n} \tag{3}$$

where n, y_j, and $\hat{y}_j$ are sample size, observed values, and predicted values of the target variable, respectively. In the case of variance of error, x_j is the residual and $\bar{x}$ is the mean of the residuals. The values of MAE, RMSE, and variance of error are always non-negative.

These metrics of model accuracy respond differently to different patterns of errors within the data set. The variance of error describes the degree of variability among the residuals, whereas RMSE and MAE summarize the magnitude of the residuals. Therefore, RMSE and MAE would not necessarily increase with variance of error. Because RMSE squares the errors before calculating their mean, large errors are given relatively higher weights. In contrast to RMSE, MAE is a linear score, meaning all residuals are weighted equally. However, MAE has the tendency to decrease faster than RMSE as sample size increases.

The similarity of MBE to MAE frequently causes it to be included in discussions about MAE. However, MBE is not a measure of accuracy. With MBE, positive and negative errors cancel each other, muting the magnitude of errors. However, in this way MBE becomes a useful indicator of model bias (Eq. (4)).

$$\mathrm{MBE} = \frac{1}{n}\sum_{j=1}^{n}\left(y_j - \hat{y}_j\right) \tag{4}$$

4.1.2. VE metrics

A commonly used measure of VE in DSM is the coefficient of determination or R^2 [40]. This metric indicates the proportion of the variance in the target variable that the model explains (Eq. (5)).

$$\mathrm{R}^2 = \frac{\sum_{j=1}^{n}\left(x_j y_j - \overline{xy}\right)^2}{\left(\sum_{j=1}^{n} x_j^2 - \bar{x}^2\right)\left(\sum_{j=1}^{n} y_j^2 - \bar{y}^2\right)} = 1 - \frac{SSRE}{SST} \tag{5}$$

where x_j and y_j are the observed and predicted values, respectively, and $\bar{x}$ and $\bar{y}$ are their respective means. In other words, R^2 is the ratio between the sum of the squared regression residual (SSRE) and the sum of the squared total residual (SST), subtracted from one. R^2 compares how much the model improves prediction over simply using the mean of the observed target variable consistently as the prediction. Using R^2 raises two major limitations: (1) it cannot determine the bias of coefficient predictions, and (2) it does not determine whether a regression model is reliable. For example, a low R^2 value might be achieved for a model with small error values (well fit models) or a high R^2 value might be achieved for a model with high error values (poorly fit models), due to the existence of a few influential and/or outlier points. This metric is dimensionless and therefore digital soil mappers can consider this metric for comparison of model performances between different target variables.

4.1.3. Other validation metrics

There are other metrics, e.g., RPD, RPIQ, and CCC, that may be useful in evaluating predictive maps. RPD and RPIQ compare standard deviation (SD) and interquartile range (IQ) to the RMSE, respectively (Eqs. (6) and 7).

$$\mathrm{RPD} = \frac{\sqrt{\frac{1}{n}\sum_{j=1}^{n}\left(x_j - \bar{x}\right)^2}}{\sqrt{\frac{1}{n}\sum_{j=1}^{n}\left(y_j - \hat{y}_j\right)^2}} = \frac{SD}{RMSE} \tag{6}$$

$$\mathrm{RPIQ} = \frac{Q3 - Q1}{\sqrt{\frac{1}{n}\sum_{j=1}^{n}\left(y_j - \hat{y}_j\right)^2}} = \frac{IQ}{RMSE} \tag{7}$$

where x_j and $\bar{x}$ are the observed values and their mean, respectively, and Q1 and Q3 are the first and third quartiles, respectively. These two metrics seek to encapsulate both accuracy and VE. RPD and RPIQ are dimensionless; therefore, they also can be compared across different target variables. Since RPD includes SD, the assumption of normal distribution of the observed values is necessary. In contrast, RPIQ instead uses IQ, which removes the need for the observed values to be normally distributed.

Although these metrics have been used for assessment of model performance [77–79], the interpretation of RPD has been arbitrary for rating model performance. For example [77–79] classified the RPD obtained from their validation tests into three different classes, weak prediction (RPD $<$ 1.4), reasonable prediction (1.4 $\leq$ RPD $\leq$ 2.0), and excellent prediction (RPD $>$ 2.0). Likewise, the RPIQ classes are also arbitrary [80], very poor model (RPIQ $<$ 1.4), fair (1.4 $\leq$ RPIQ $<$ 1.7), good model (1.7 $\leq$ RPIQ $<$ 2.0), very good models (2.0 $\leq$ RPIQ $\leq$ 2.5), and excellent models (RPIQ $>$ 2.5).

Another statistic that has been used as a measure of model performance is Lin's concordance correlation coefficient (CCC). This metric was designed to assess the agreement between two measurements of the same variable [81]. However, previous studies of DSM have considered CCC to evaluate the performance of digital maps by evaluating the agreement between the observed and predicted values [82,83]. In this context, the model predicted value could be considered an alternative method of measurement, even though it was not a direct measurement. CCC has been assigned the term p_c (Eq. (8)).

$$\mathrm{CCC} = p_c = \frac{2p\,\sigma_x\,\sigma_y}{\sigma_x^2 + \sigma_y^2 + (\mu_x + \mu_y)^2} \tag{8}$$

where p is the Pearson correlation coefficient between the two measurements, σ_x^2 and σ_y^2 are the corresponding variances, and μ_x and μ_y are the means of the two measurement methods. CCC assesses both precision and accuracy of the prediction. In this case, precision refers to the spread of points around the regression line (correlation coefficient) for the predicted versus observed. The accuracy refers to the correspondence between that regression line and the perfect line (1:1 line). The range of CCC is between −1 and 1, with perfect agreement at 1. The main difference between the Pearson correlation coefficient and CCC is that the correlation coefficient can handle the prediction of variance being biased. However, the correlation coefficient fails to capture bias in the model prediction, which would be expressed by deviations from the 1:1 line. In this way, CCC would be a more appropriate evaluation than R^2 for the correspondence between observed and predicted values.

4.2. Comparing uncertainty metrics

To compare the response of different uncertainty metrics to sample size and magnitude of residuals, the goodness of fit for a single MLR model was evaluated on four different subsets of the data used in the test cases for sample size and computation time. Dataset A contained 200 sample points. Dataset B was a subset of dataset A, containing only 50 points. To evaluate the effect of error magnitudes, dataset B was divided into the 25 points with the largest residuals (dataset C) and the 25 points with smallest residuals (dataset D).

The largest difference between RMSE and MAE occurred with dataset A, where the RMSE was much larger than MAE (Table 6). RPD, RPIQ, and CCC were also affected by the quantity of samples because they include the calculation of RMSE and variances. However, the effect of sample size on these metrics was not remarkable compared to RMSE and MAE.

The error magnitude is another factor that can influence the results of these metrics. In this case study, the influence of error magnitude on these metrics was tested by holding the sample size constant while selecting for different magnitudes of error in datasets C and D. The variance of error in dataset C was around four times larger than the variance of error in dataset D. As a result, RMSE and MAE increased when the error magnitudes were larger, while there was no remarkable change in RPD, RPIQ, or CCC values. The RMSE and MAE for dataset C were almost one and a half and two times larger, respectively, than the RMSE and MAE for dataset D. The differences between the other validation metrics, e.g., RPD and RPIQ, in datasets C and D were much less. However, the smaller residuals of dataset D resulted in the value of CCC increasing by 50% compared to dataset C.

Any single metric supplies a certain perspective on the model uncertainty. Therefore, a combination of metrics can provide a more complete description of the model performance [84]. However, choosing the most appropriate uncertainty metric would help users to draw conclusions simply and scientifically.

4.3. Model and map evaluation

To assess the model performance of digital soil maps, some studies investigate map performance based on two independent datasets, a training dataset and a separate validation dataset [55,64,85]. Others use k-fold CV to assess the map performance [58,59,71]. CV is a useful strategy to maximize the quantity of points in the training dataset but does not measure the uncertainty of the model that would be trained on the whole set of sample points. CV addresses the question of how much the results would change if different points had been used for training. A useful term for this type of validation

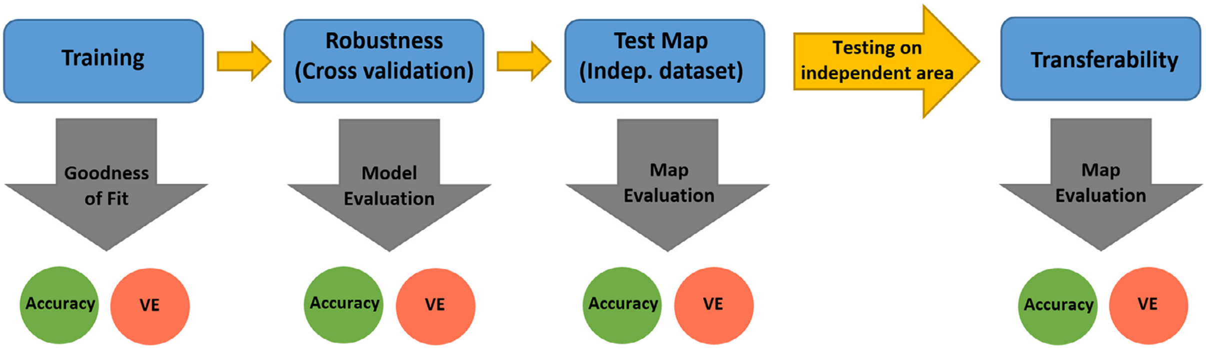

Fig. 5. Flowchart for assessing the model and map performance in DSM. Four main steps are recommended: (1) training the model with one dataset (goodness of fit), (2) testing the model performance with CV (robustness), (3) testing the map validation inside the same geographic extent with an independent dataset, and (4) testing the transferability of the model in a different geographic area with a second independent dataset.
VE, Variation Explained.

Table 7
Major factors for selecting suitable ML algorithm(s) on DSM.

Algorithm(s)	Quantity of Hyperparameters	Not Sensitive to Data Size	Covariate Selection*	Low Computation Time	Interpretability
MLR	0	×	×	√	√
KNN	1	√	×	√	×
SVR	2	√	×	√	×
Cubist	2	√	√	√	√
RF	3	√	√	×	×†
ANN	8	×	×	×	×

ANN, Artificial Neural Networks; KNN, K-Nearest Neighbor; MLR, Multiple Linear Regression; RF, Random Forest; SVR, Support Vector Regression.
* The X symbol (×) means that ML algorithms use all covariates and the check mark (√) means that the algorithm has the ability to use covariate selection to reduce the quantity of variables that need to be processed by the algorithm.
† RF is not completely uninterpretable. The model produced by RF is semi-black box.

is model robustness because it tests dependency of the results on which points were selected for training. A large decrease in performance metrics between the goodness of fit (training) and the CV would indicate the models being generated are not robust. However, if the proportion of sample points set aside in each fold of the CV is small, then the test would become less sensitive to any issues with model robustness.

To fully evaluate the prediction performance of a model, we recommend three levels of validation beyond the training assessment (Fig. 5). At the first level, after the model is trained on the primary dataset, CV is run to test robustness. This is the minimal evaluation that every digital soil mapping study should report. While other studies may have the resources to perform higher levels of validation, reporting this step would facilitate comparison with other studies that do not have sufficient sample points to conduct higher level validation.

The second level of evaluation is an independent validation using points within the same geographic extent of the training points but not used in the first level of model evaluation. This step in the evaluation truly tests the predictive power of the model. Because the model is changed with every fold of CV, that test does not directly evaluate the performance of the predictions that will become the map product.

The ideal, ultimate goal is to produce a generalized model that has predictive power outside of the area where the model was trained. This concept is complicated and may be less applicable in situations where the study area has a large extent. However, how specific the model developed is to the study area is an important question for the advancement of science. For example, when a digital soil map of a field is produced, it would be useful to know if the model is transferable to an adjacent or other nearby field. To accomplish this level of model evaluation, a second independent set of points located beyond the geographic extent of the training dataset would be held in reserve and then the model's ability to predict those external locations would be tested in a separate, more challenging round of independent validation.

5. Synthesis of important actions

To select appropriate ML algorithms for DSM studies, consideration of the limitations and strengths of different algorithms is necessary (Table 7). Since there is a positive relationship between the quantity of hyperparameters and computation time, algorithms with fewer hyperparameters train faster. For this reason, MLR, KNN, Cubist, and SVR provide results in less time than ANN and RF.

When a data set consisting of more than a thousand soil samples is available and computation time is not a concern, ANN is likely to provide the best predictions of a target variable. When the data set contains less than a thousand soil samples, ANN is likely to provide unstable results and a different ML algorithm should be considered. Cubist, KNN, RF, and SVR, can provide stable results with data sets containing less than a hundred soil samples.

If interpretability of the model is important to the project's goals, e.g., identifying unknown relationships between covariates and soil properties, MLR, Cubist, and to some degree RF, would be useful algorithms. An interpretable model could suggest opportunities for investigating soil formation processes when selected covariates indicate correlations not previously recognized in soil science. However, MLR is extremely vulnerable to both overfitting and underfitting and it cannot produce results as strong as those of Cubist and RF.

For evaluating model performance, we propose a three-tier framework whereby in ideal conditions all levels of evaluation would be conducted. These evaluation levels are: (1) assess goodness of fit and CV on the training data set, (2) test prediction performance on an independent set of points within the same geographic area as the training data, and (3) test the transferability of the model by performing a second independent validation on a separate set of points located outside of the training data's geographic extent. We recognize that it may not always be practical to use all three tiers of this framework, but we encourage researchers to consider them in their experimental design. Whenever higher levels of evaluation are possible, the lower levels of evaluation should still be performed and reported to maintain comparability with studies that were not able to conduct higher levels of evaluation.

5. Conclusions

Understanding the strengths and weaknesses of different ML algorithms is critical for selecting the most appropriate ML algorithm for a particular DSM task. ANN produces strong results for predicting non-linear patterns. RF is faster than ANN and its results also tend to be strong. RF and Cubist overcome the weakness of ANN's sensitivity to small datasets and being an entirely black box model. However, Cubist produces results that are as strong as RF, but with more direct information about the model used to make predictions.

The evaluation of results should be based on two different but equally important types of metrics: (1) accuracy and (2) VE. The classic metrics for these are RMSE and R^2. RMSE provides a useful measure of accuracy, while R^2 indicates the degree of variation explained by the model. However, R^2 does not detect prediction bias as the fitted line between observed and predicted values may not be 1:1. For this reason, while R^2 is suitable for evaluating a model's goodness of fit for the training data, CCC is a more suitable validation metric. RMSE is a useful metric for assessing the map accuracy in the units of the target variable, which can be the most important test for certain applications. CCC and R^2, on the other hand, assesses the map's ability match the spatial pattern of the target variable. An advantage for CCC and R^2 is that by being dimensionless, comparisons between a wider variety of DSM projects are easier than with RMSE. Although evaluating models with other metrics may be useful, the fundamental metrics of RMSE, CCC, and R^2 should still be reported for comparisons with other studies.

To assess the performance of a model and the resulting predicted map, we recommend three tiers of evaluation. At the first level, the robustness of the model should be tested by training on the primary dataset and running CV. For the second level of validation, an independent validation inside the same geographic extent is required to test the predictive power of the model. And finally, another independent validation test to evaluate the transferability of the model outside the area where the model is trained would be beneficial. We recognize that not every DSM project will have the resources to evaluate all three tiers. Therefore, every DSM study should report the results of lower level tiers of evaluation to maintain comparability with other research efforts.

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